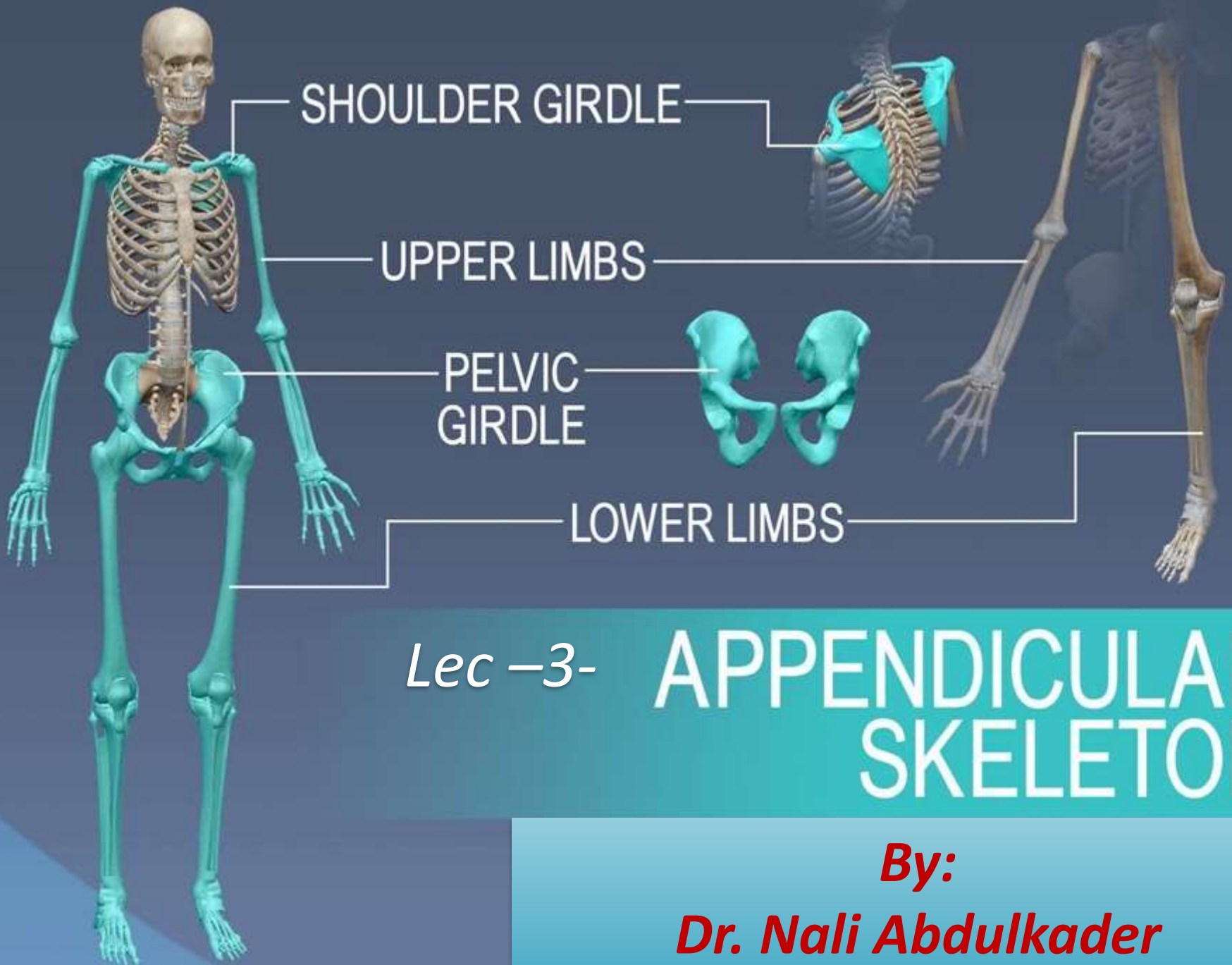




Lec –3-

APPENDICULAR SKELETON

By:

Dr. Nali Abdulkader

An anatomical diagram of the human upper limb skeleton. The humeri (upper arm bones) are colored yellow, the radius and ulna (forearm bones) are colored red, and the carpal, metacarpal, and phalanx bones of the hand are colored green. The diagram shows the bones in a frontal view, with the right arm slightly raised and the left arm hanging down. The rib cage and spine are visible in the background.

Lec: 3
Bones of the
Upper limb

By:
Dr. Nali Abdulkader

Upper limb

The upper limb is located in the lateral aspect of lower portion of the neck and thoracic wall.

It is suspended from the trunk by muscle and small skeletal articulation between the clavicle and the sternum.



Parts of the upper limb

❖ The upper limb is divided in to

➤ *Shoulder region*

Shoulder joint

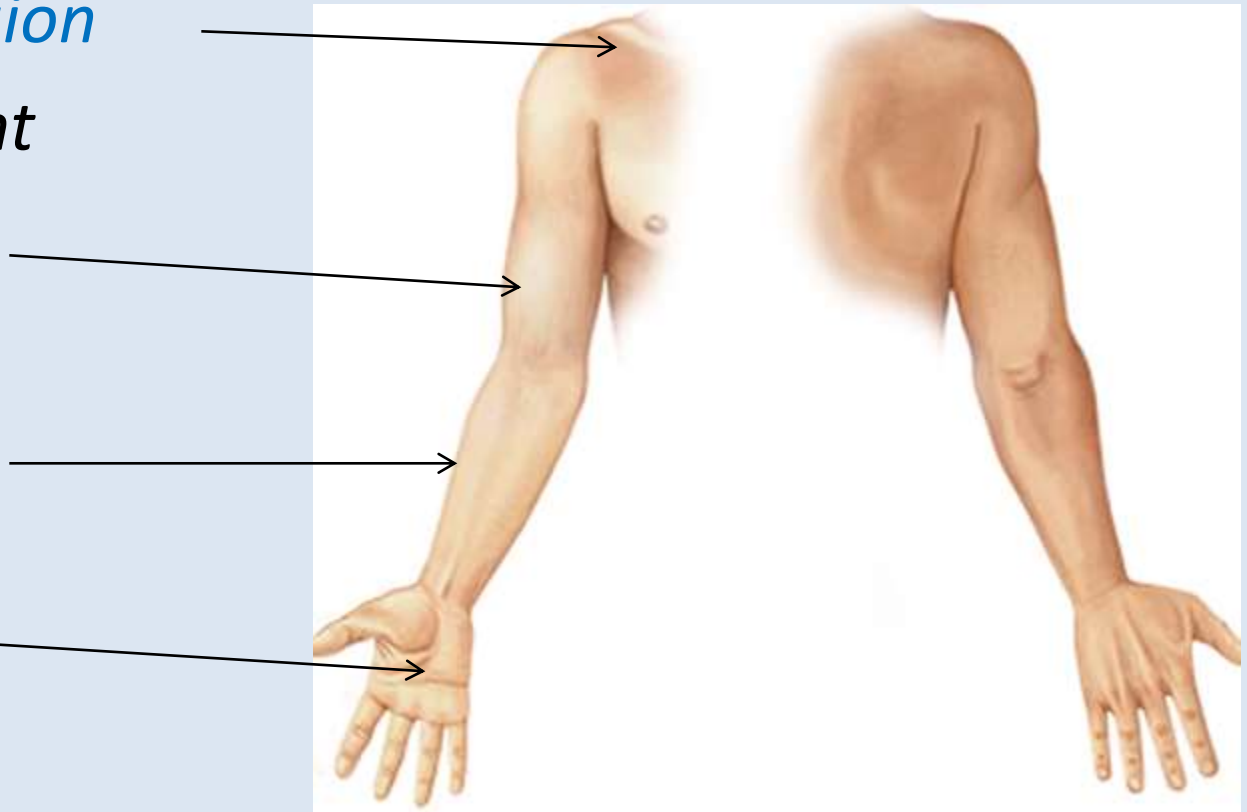
➤ *The arm*

Elbow joint

➤ *The forearm*

Wrist joint

➤ *Hand*



Skeleton of the Upper limb

❖ Shoulder girdle :

- Clavicle and Scapula

❖ Arm:

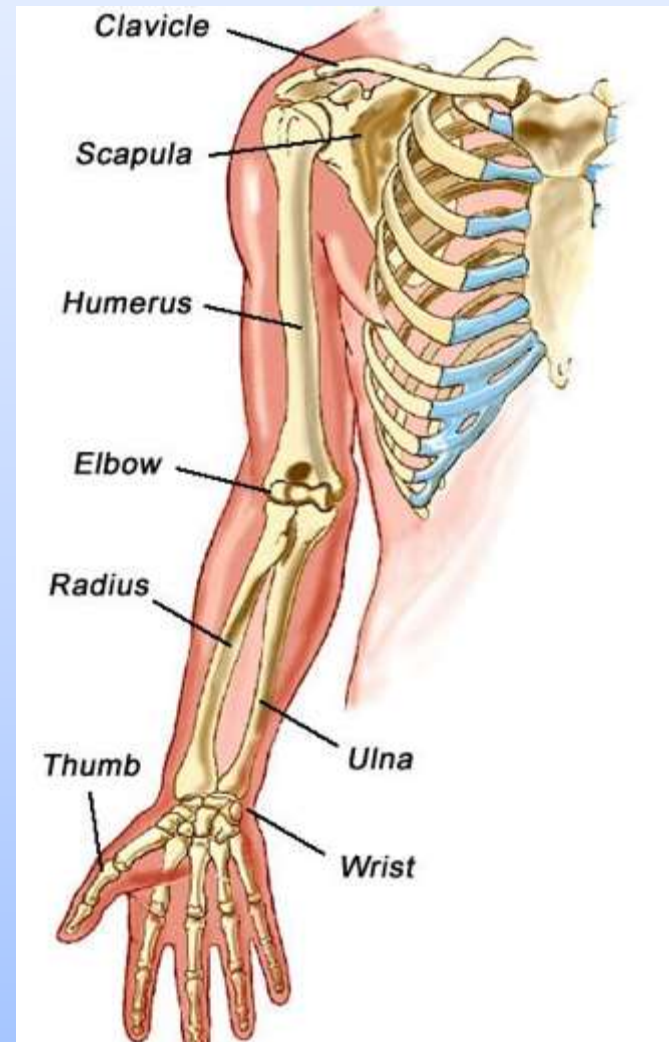
- Humerus

❖ Forearm :

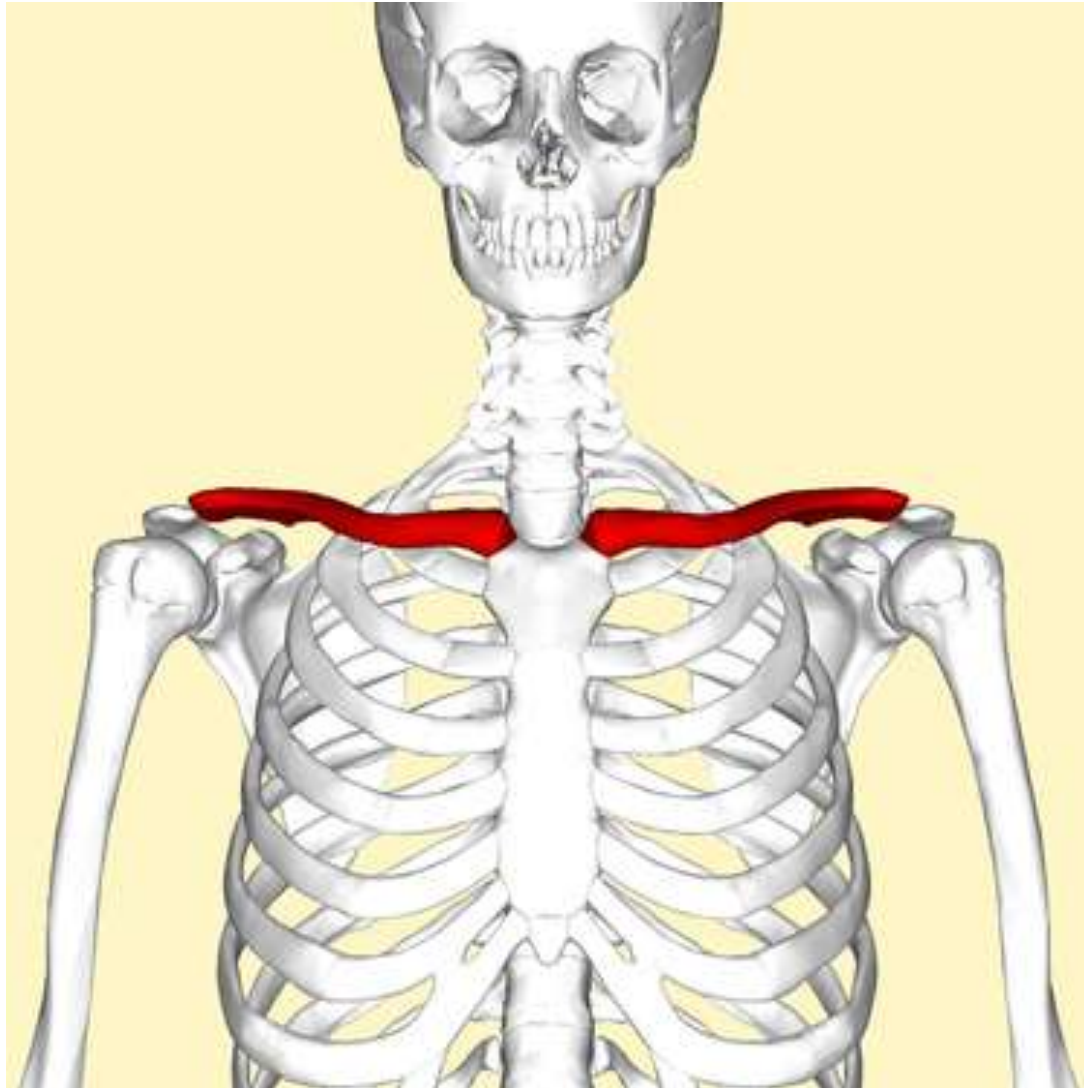
- Radius and Ulna

❖ Bone of the hand:

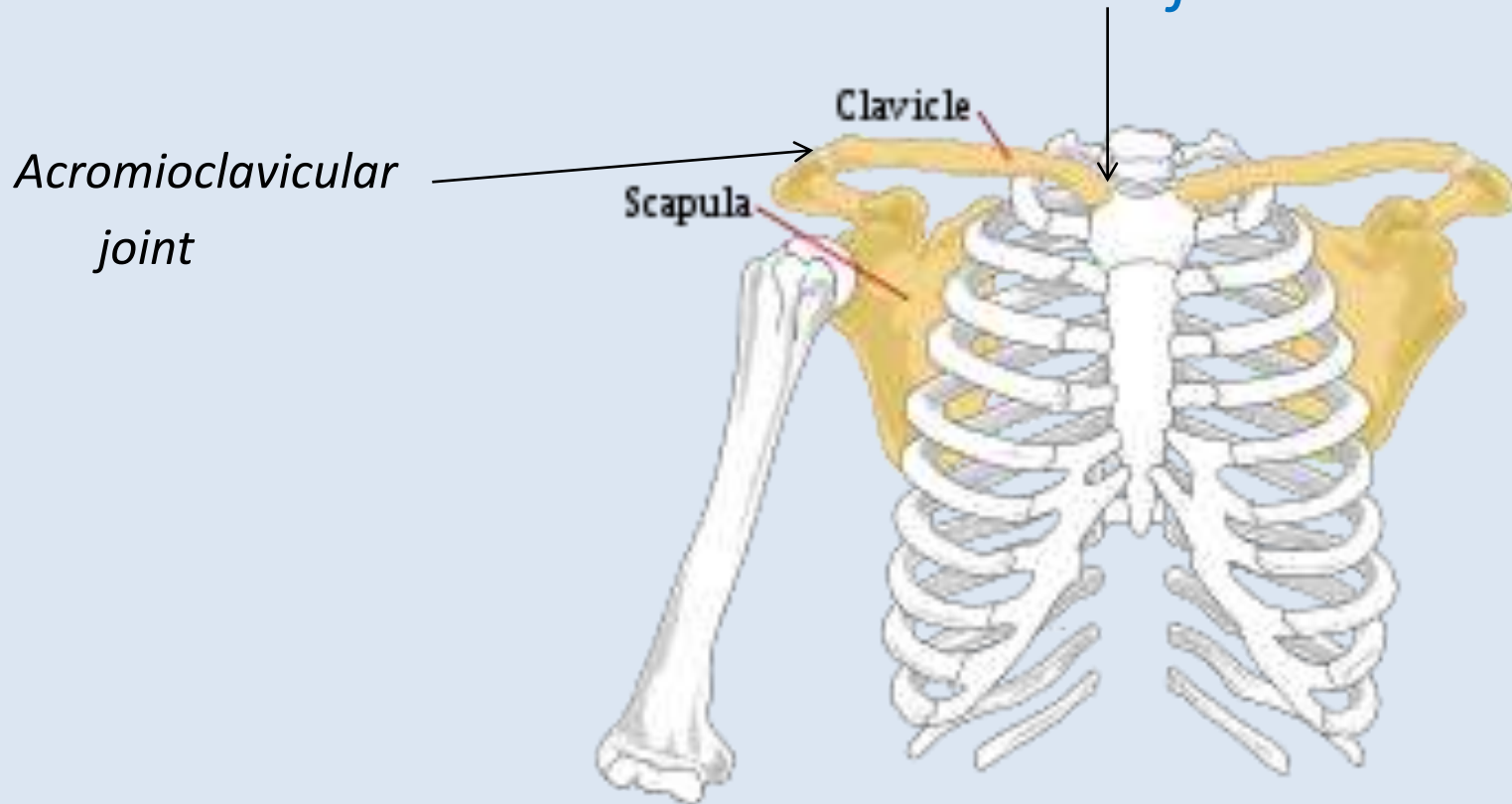
- Carpal, Metacarpal and Phalanges



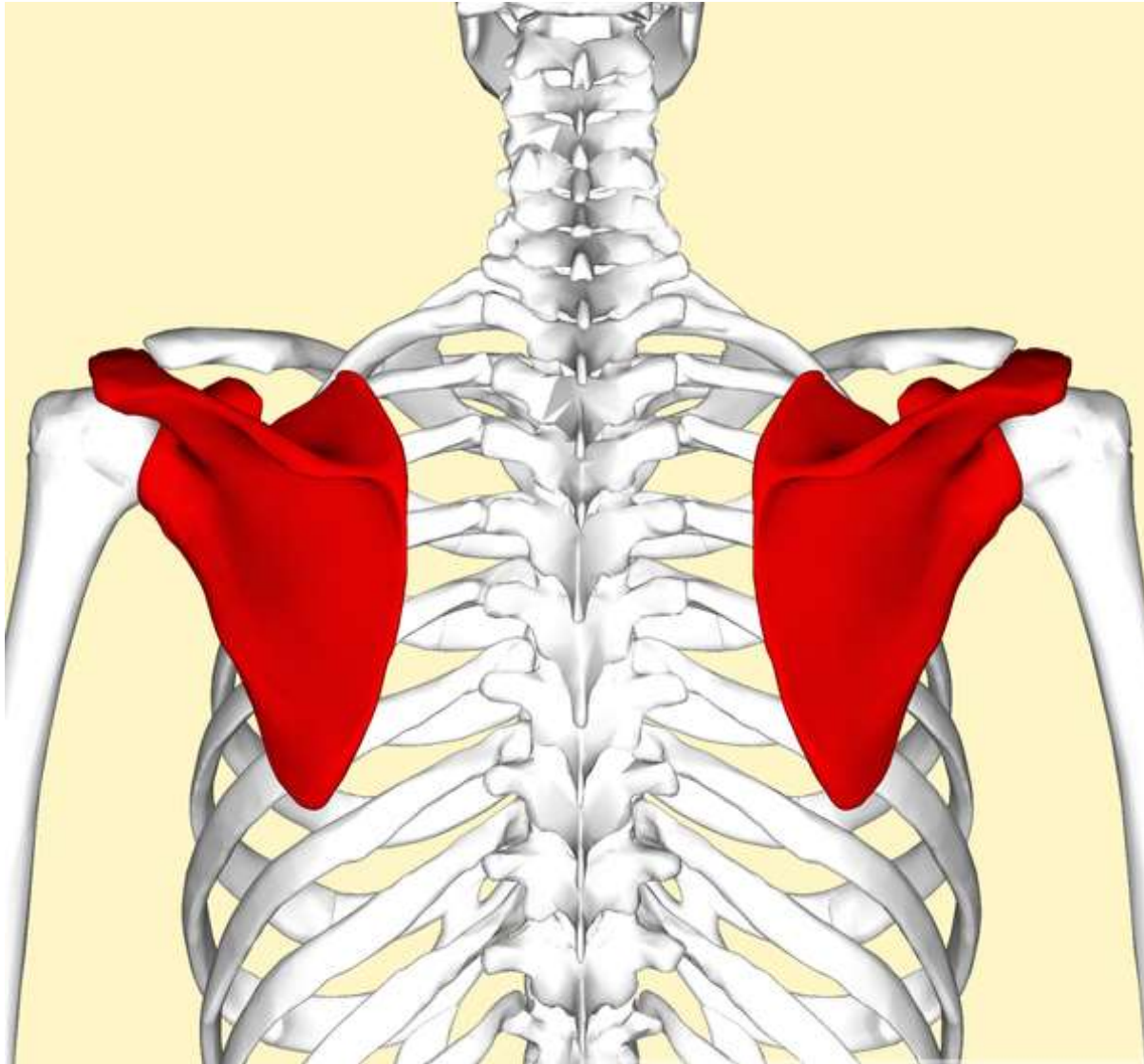
Clavicle



- ❖ **Lateral end** of the clavicle called acromial end because it articulates with acromial of scapula to form *Acromioclavicular joint*
- ❖ **Medial end** also called sternal end because it articulates with sternum to form *Sternoclavicular joint*



Scapula



Scapulas (holder blade)

- Is flat, postrolateral aspect of the thorax
- Over lying 2nd to 7th rib
- It has two surfaces, three borders, three angles and three process



Surfaces of the scapula

A. Costal surface

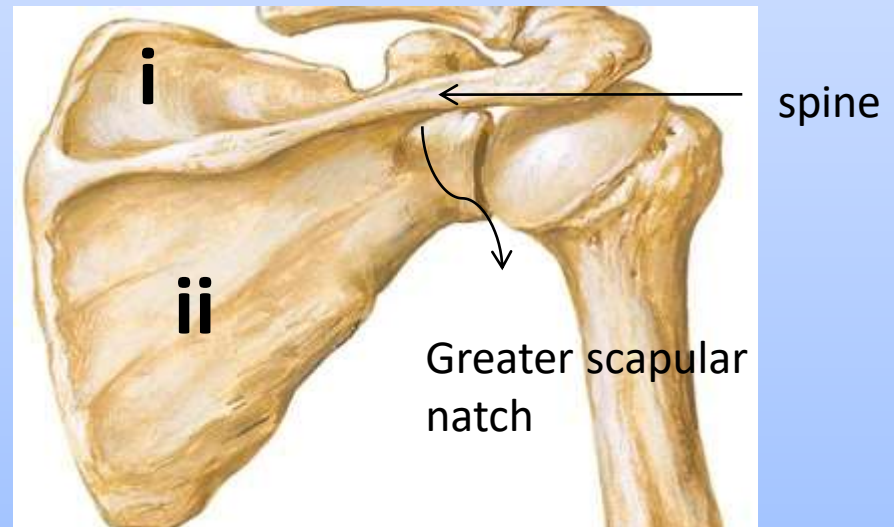
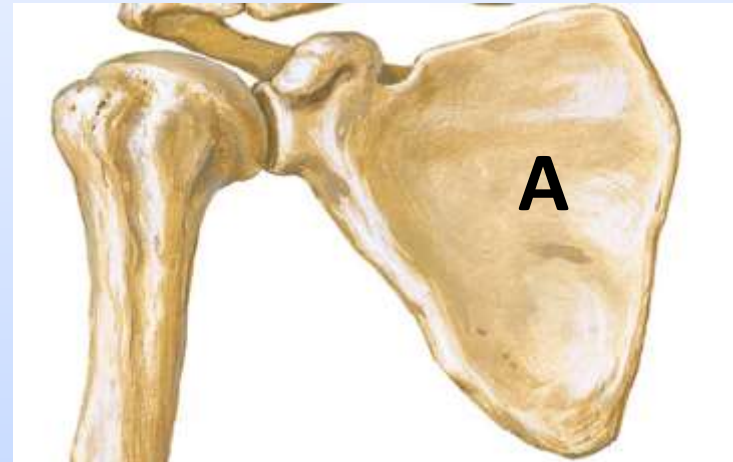
(subscapular fossa) its marked by three longitudinal ridge

B. Dorsal surface

divided by spine into

- i. supra and
- ii. infra spinatus fossa

The two fossa conneted by greater scapular notch



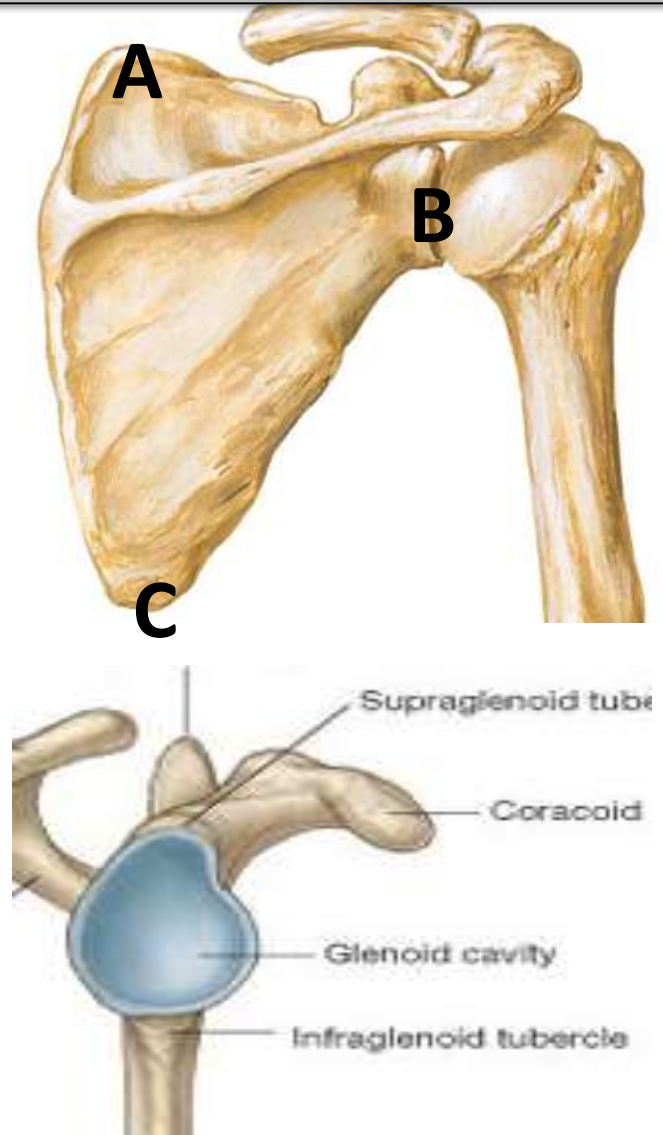
Angles

❖ *It has three angles*

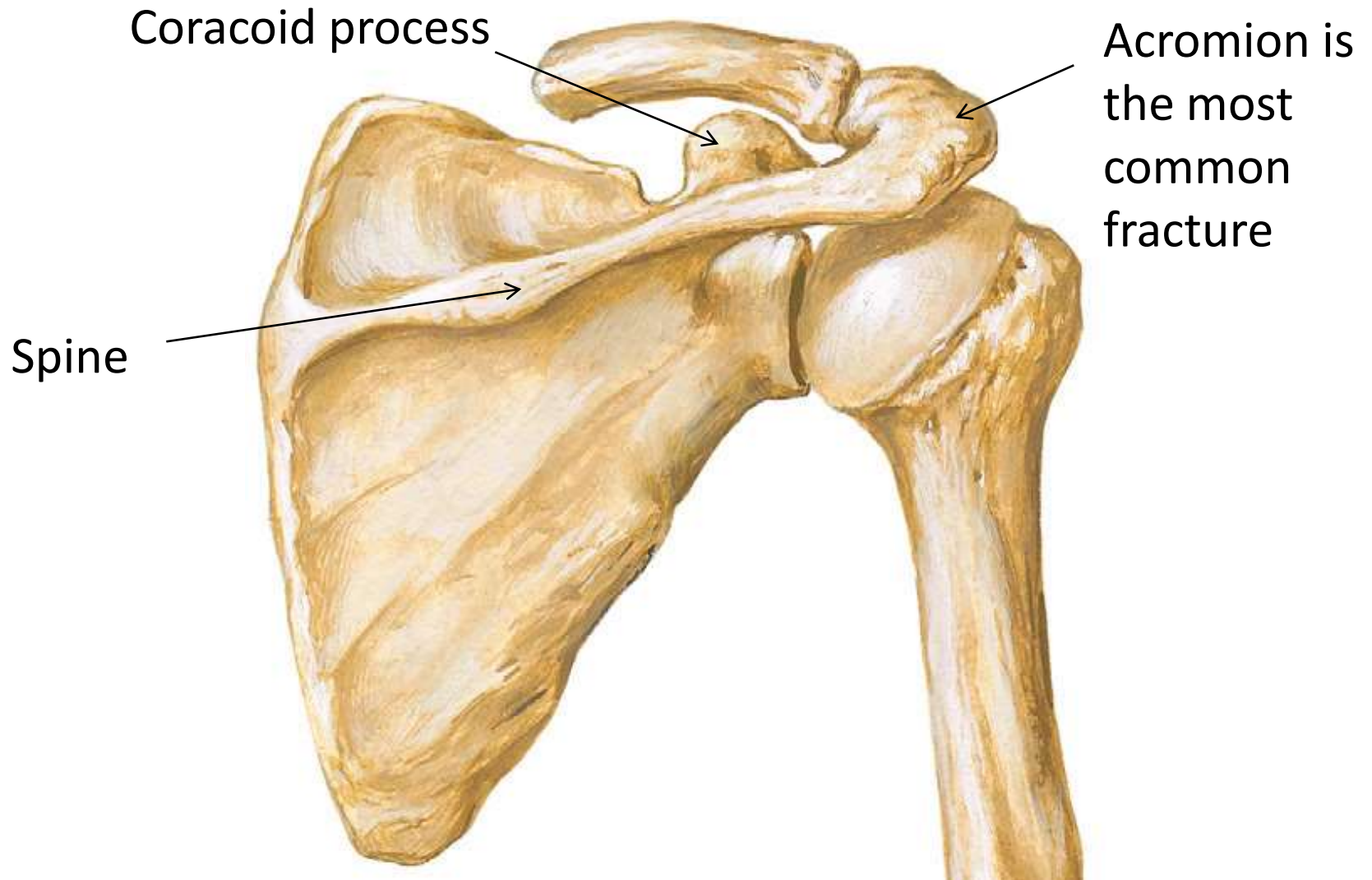
A. Superior

B. Lateral shallow ,comma shaped called glenoid cavity which has infra and supra glenoid tubercle

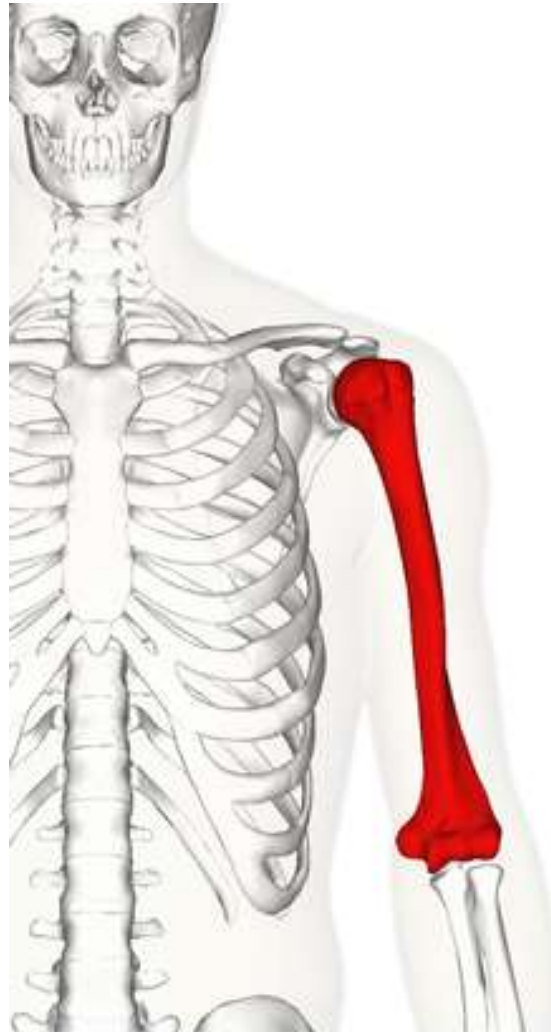
C. Inferior



The processes

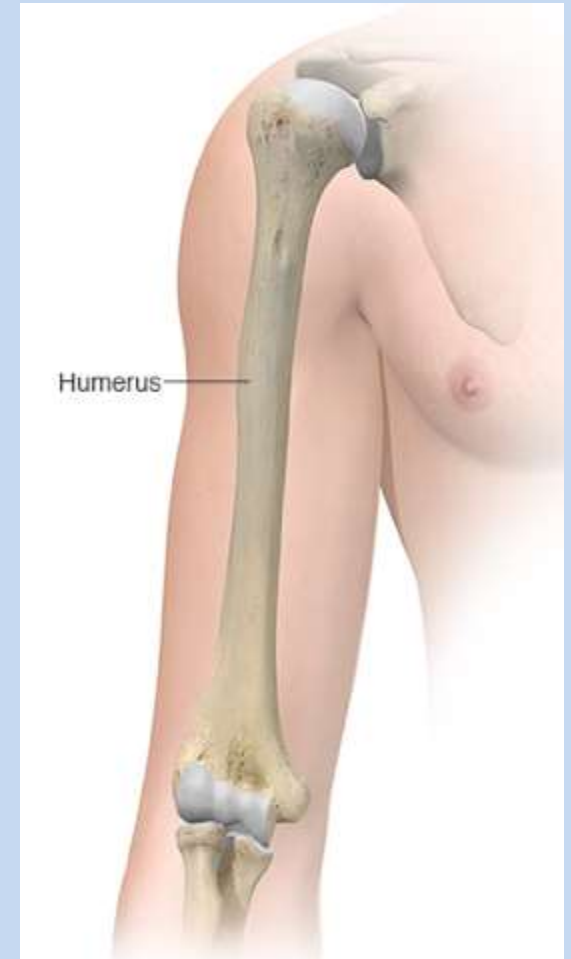


Humerus



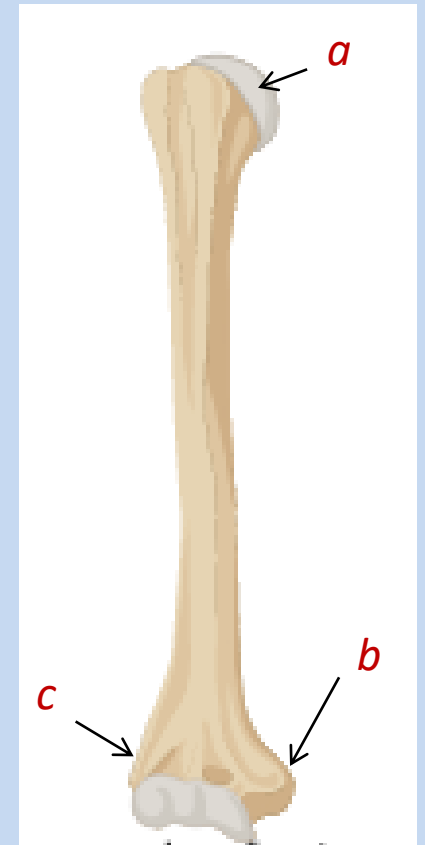
Humerus

- ❖ The **humerus** is a long bone of the upper limb, which extends from the shoulder to the elbow.
- ❖ Like other long bones it has two ends and shaft



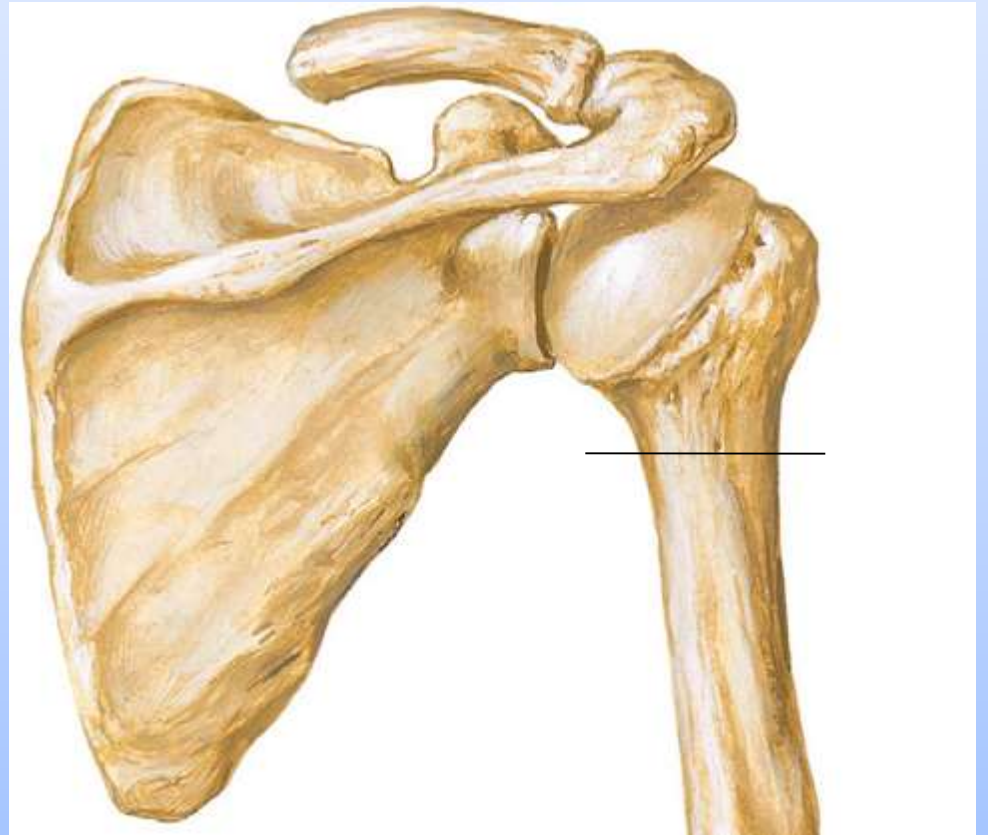
Humerus

- ❖ The proximal aspect of the humerus has the a. head, articulates with the glenoid cavity of the scapula, forming the shoulder joint.
 - ❖ Distally, it articulates with Radius and Ulna , forming the elbow joint
- Distal end also has two processes called
- b. Medial and c. lateral epicondyles



Surgical neck of humerus

❖ Is the narrowest part just distal to the head



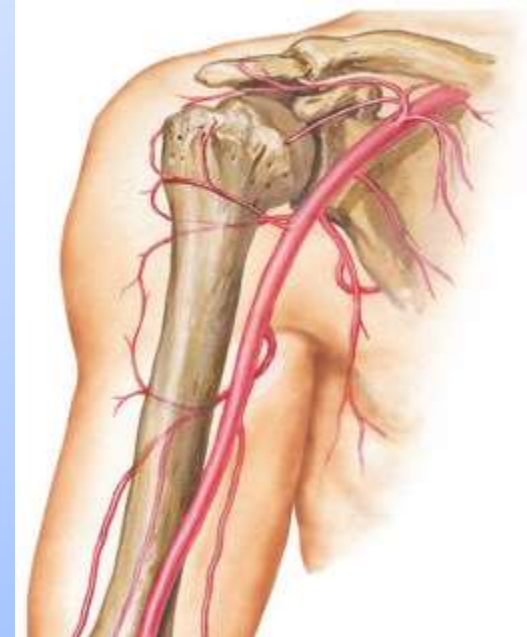
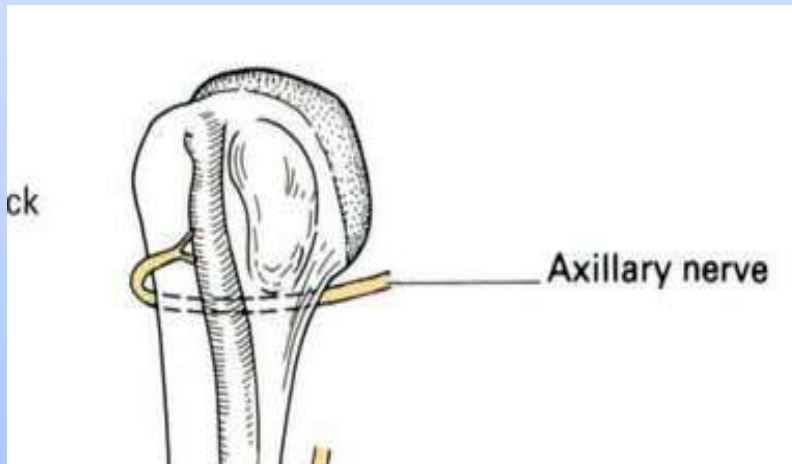
Fracture of surgical neck

- The surgical neck is the most common site for fracture specially in elder people with osteoporosis
- Due to direct fall on the lateral aspect of the shoulder



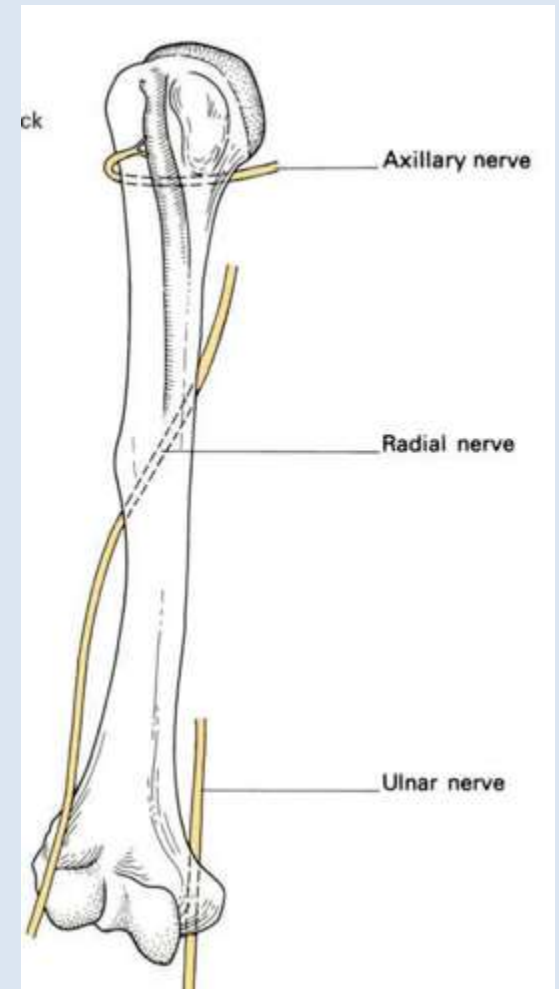
Fracture of surgical neck

- Fracture of the surgical head of the humerus may cause injure of the axillary nerve and rupture of profundus brachial artery



Relations of nerves with the humerus

- **Surgical neck:** axillary nerve.
- **Radial groove:** radial nerve.
- **Distal end of humerus:** median nerve.
- **Medial epicondyle:** ulnar nerve.
- These nerves may be injured when the associated part of the humerus is fractured.

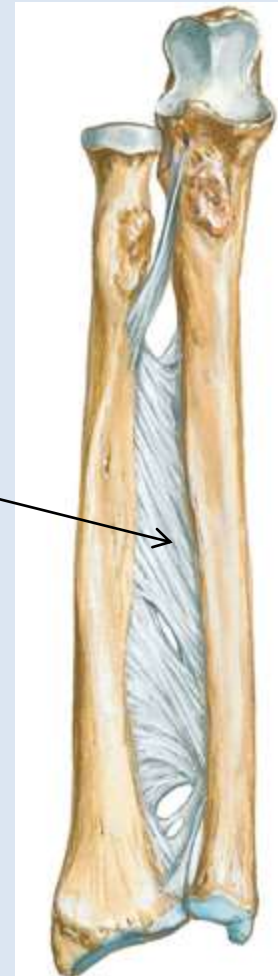


Forearm

❖ The bones of the forearm
are radius and ulna

Two bone connect with each other

By **interosseous membrane**



The radius

- ❖ The radius is the lateral bone of forearm
- ❖ Its shorter of the two forearm
- ❖ It has upper and lower ends and the shaft



It forms 4 joints of upper limb

1- Elbow, 2- superior and 3- inferior radioulnar joints and 4- wrist joint

The upper end of the radius

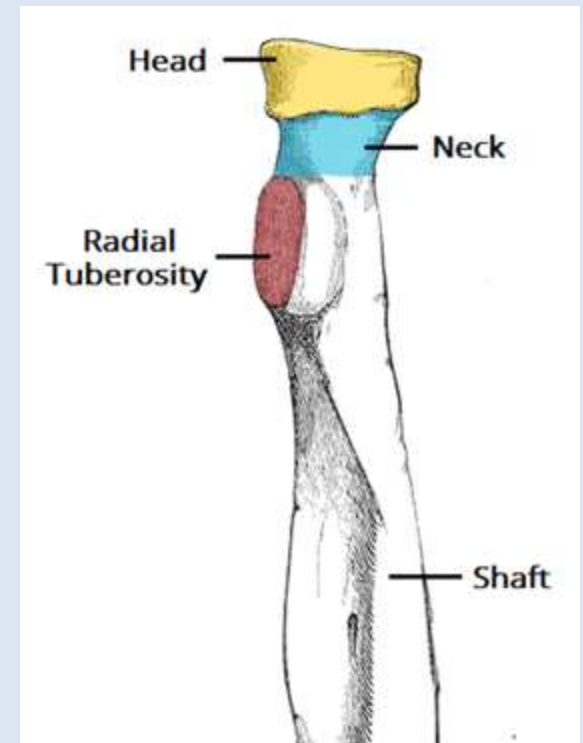
❖ The proximal end of the radius consists of

➤ **Head** is thick and disc shaped

➤ **Neck** is short and narrow

➤ **Radial tuberosity**

❖ The head articulates superiorly with the humerus to form the elbow and medially with ulna to form inferior radioulnar joint



The distal end of the radius

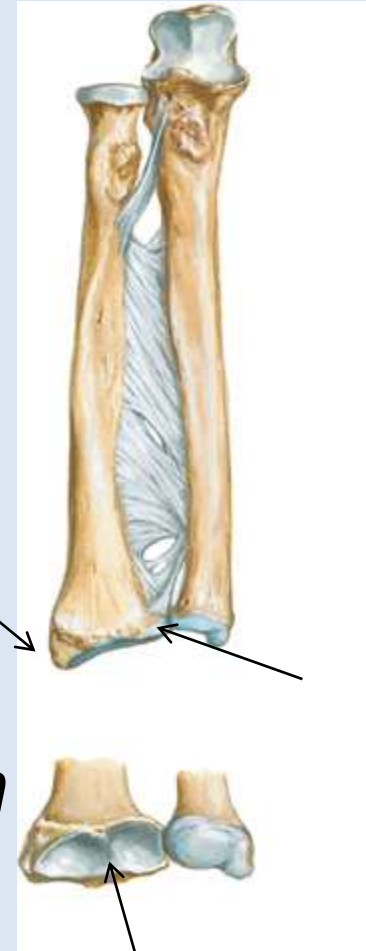
❖ *It articulates distally with the Ulna to form inferio-radioulnar Joint*

*Laterally it has styloid process
Of radius*

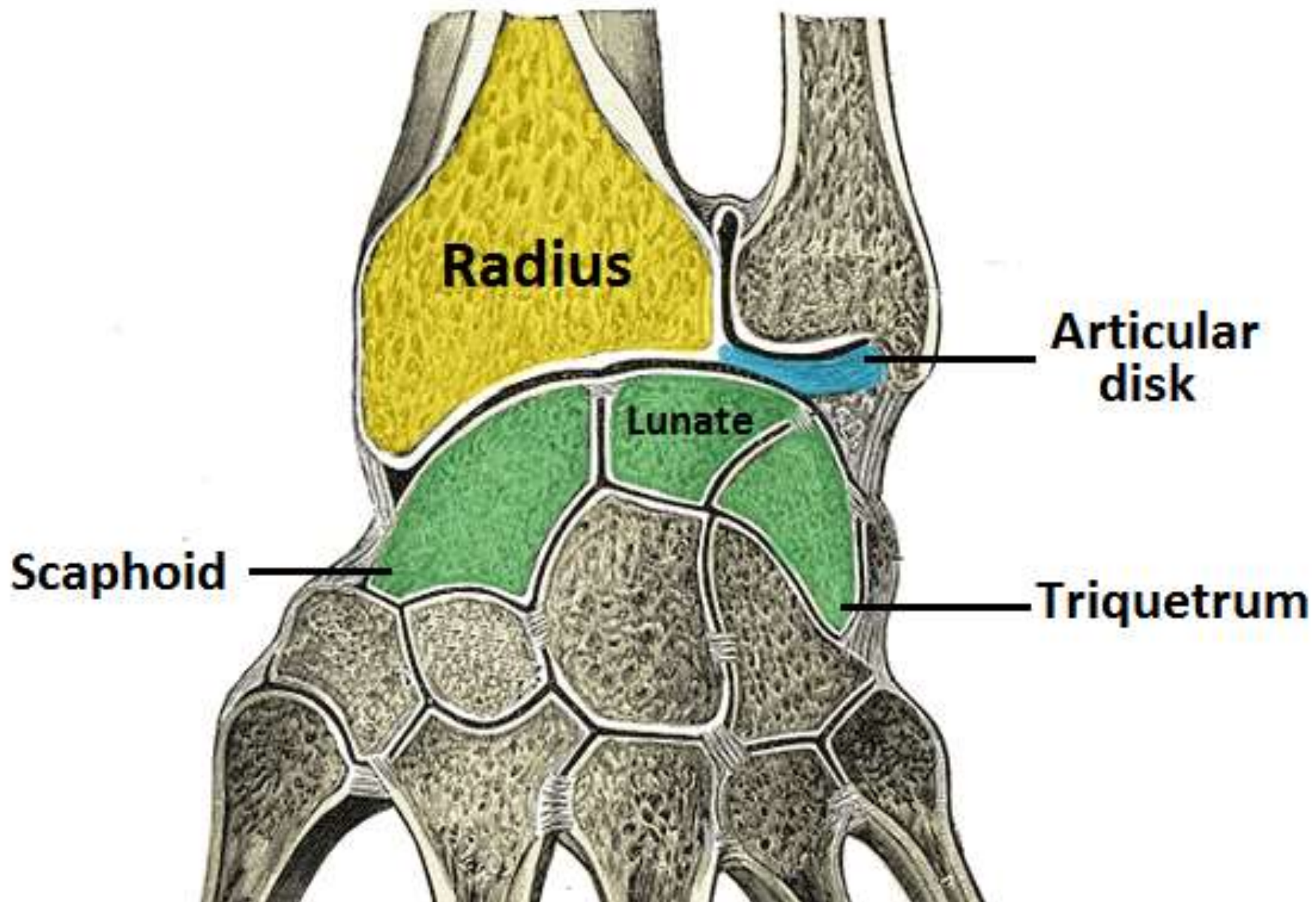
Inferior surface of radius

*For articulates with carpal to form
Wrist joint*

(styloid
process)



Inferior surface



The radius

- The proximal end:** *a head..., neck ..and tuberosity*
- The shaft:** *terosseous border*
- The distal end:** *styloid process*

The wrist joint is made up between radius and two carpal bones; the ulna does not share in formation of wrist joint .

The ulna

- The ulna is the medial bone of forearm
- Articulation
- It has upper and lower end and shaft



The proximal end

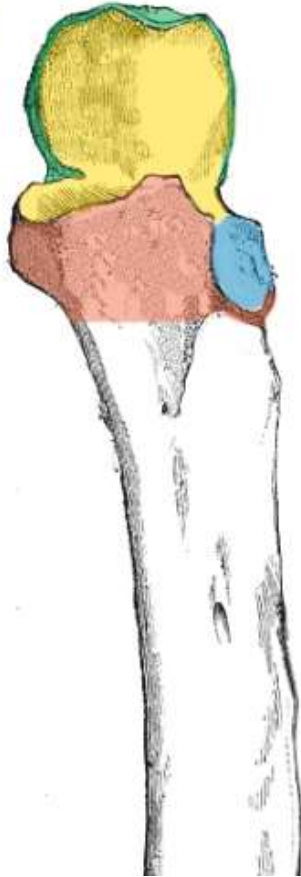
❖ *The proximal end has*

Trochlear notch for articulation with
Trochlear of humerus

Radial notch for articulation with the
head of radius



Anterior:



Lateral:



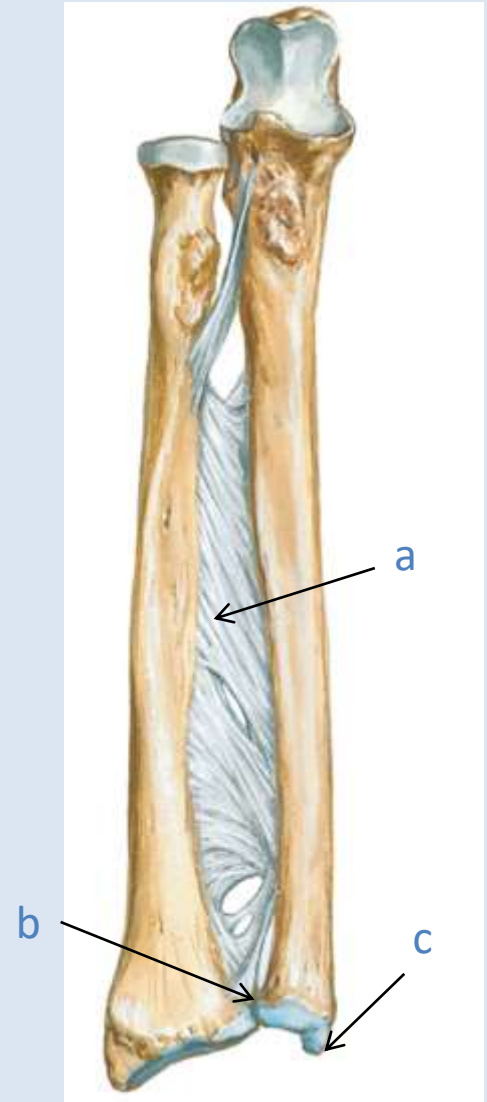
-  Trochlear Notch
-  Radial Notch

❖ The shaft of the ulna which articulates with the shaft of the radius by **a-** interosseous membrane

❖ **The lower end of the ulna** has

b- Head

c- Styloid process



Skeleton of the hand

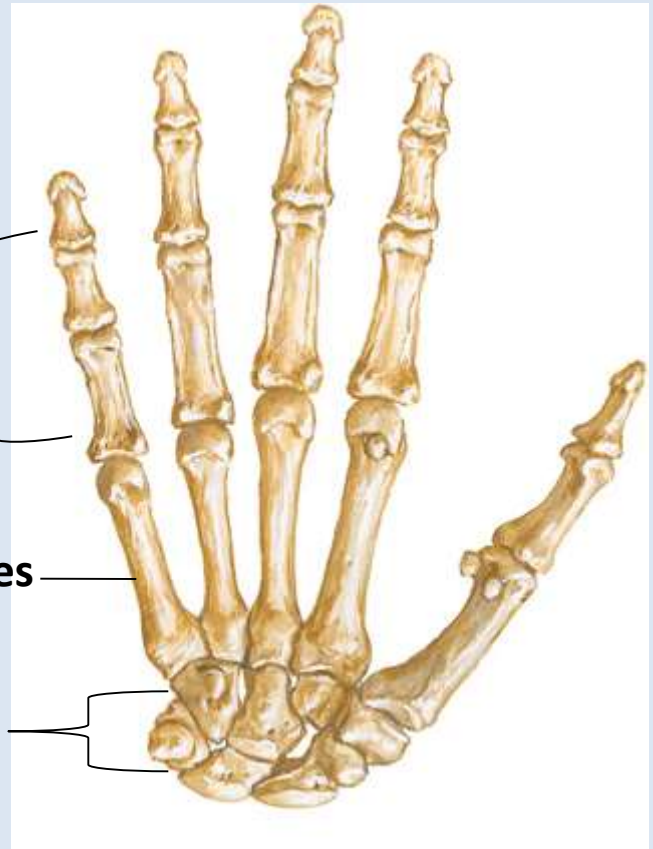
❖ include three parts:

- the carpal bones,
- the metacarpal bones
- and phalanges

phalanges

Metacarpal bones

Carpal bones



Carpal bones

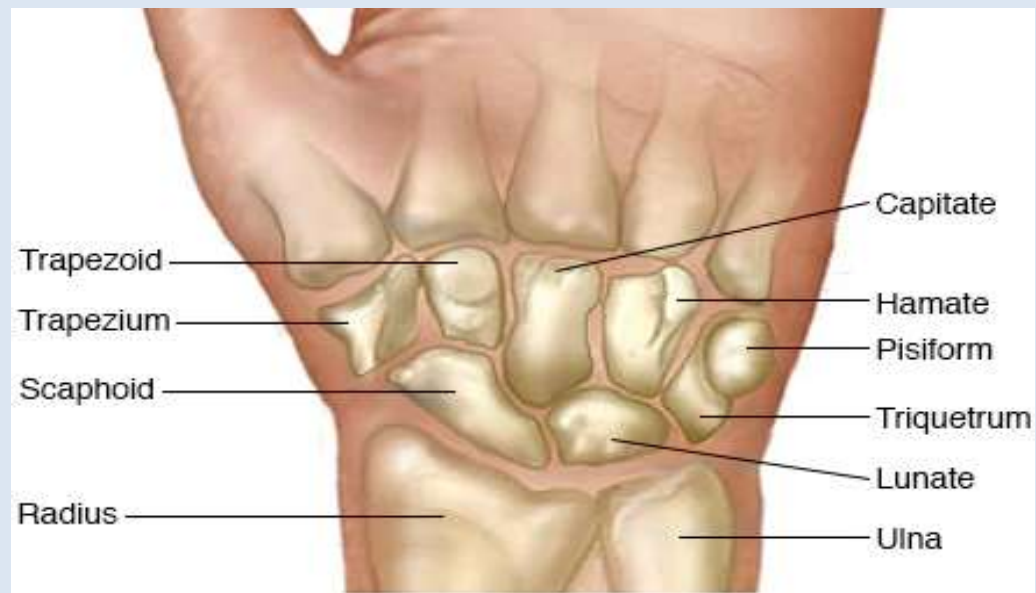
- Its 8 bones, two rows

❖ Proximal row

1- Scaphoid 2- Lunate. 3- Triquetral & 4-Pisiform

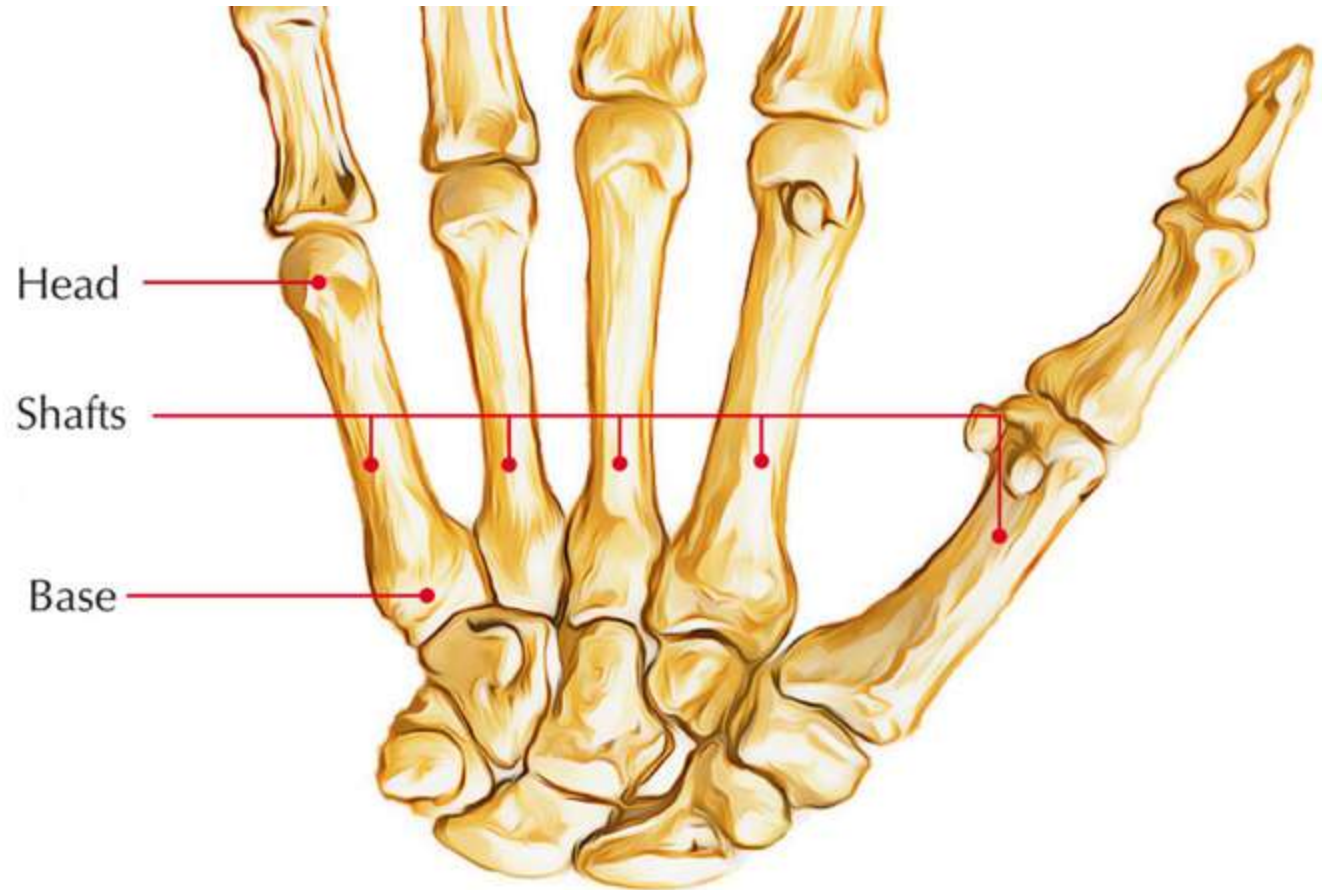
❖ Distal row

5- Trapezium. 6- Trapezoid. 7- Capitate. 8- hamate



Metacarpals

Metacarpal bones



phalanges

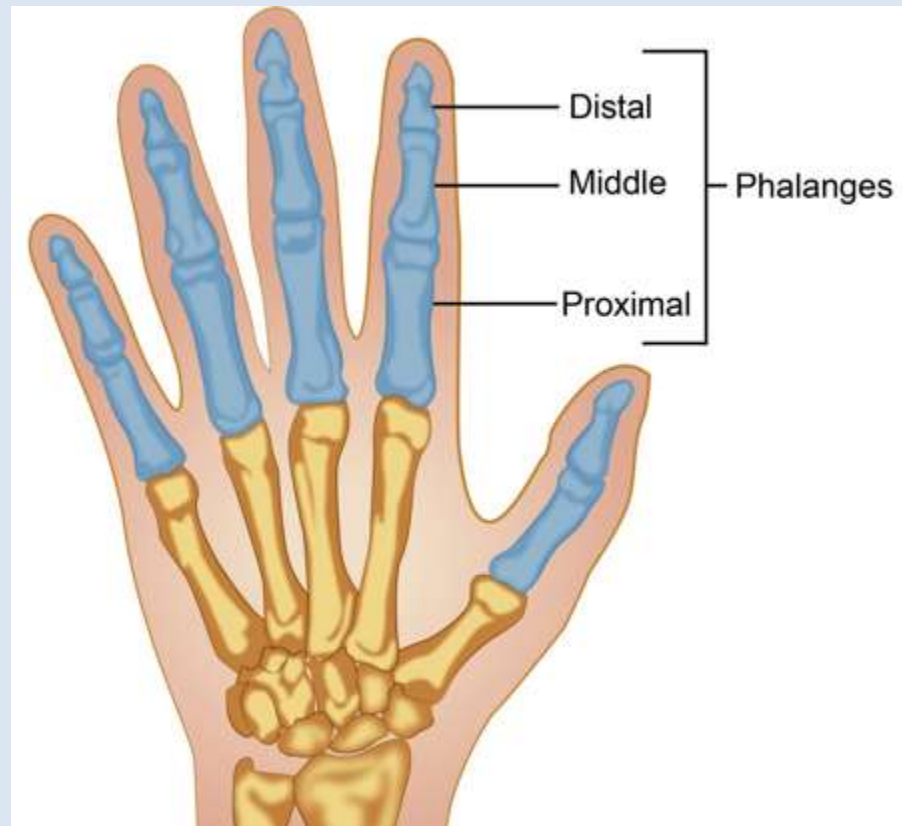
❖ There are 14 phalanges in each hand

Each digit has three phalanx :

- proximal,
- middle and distal

❖ Each phalanx

- has base and
- distal head, and
- shaft(body)

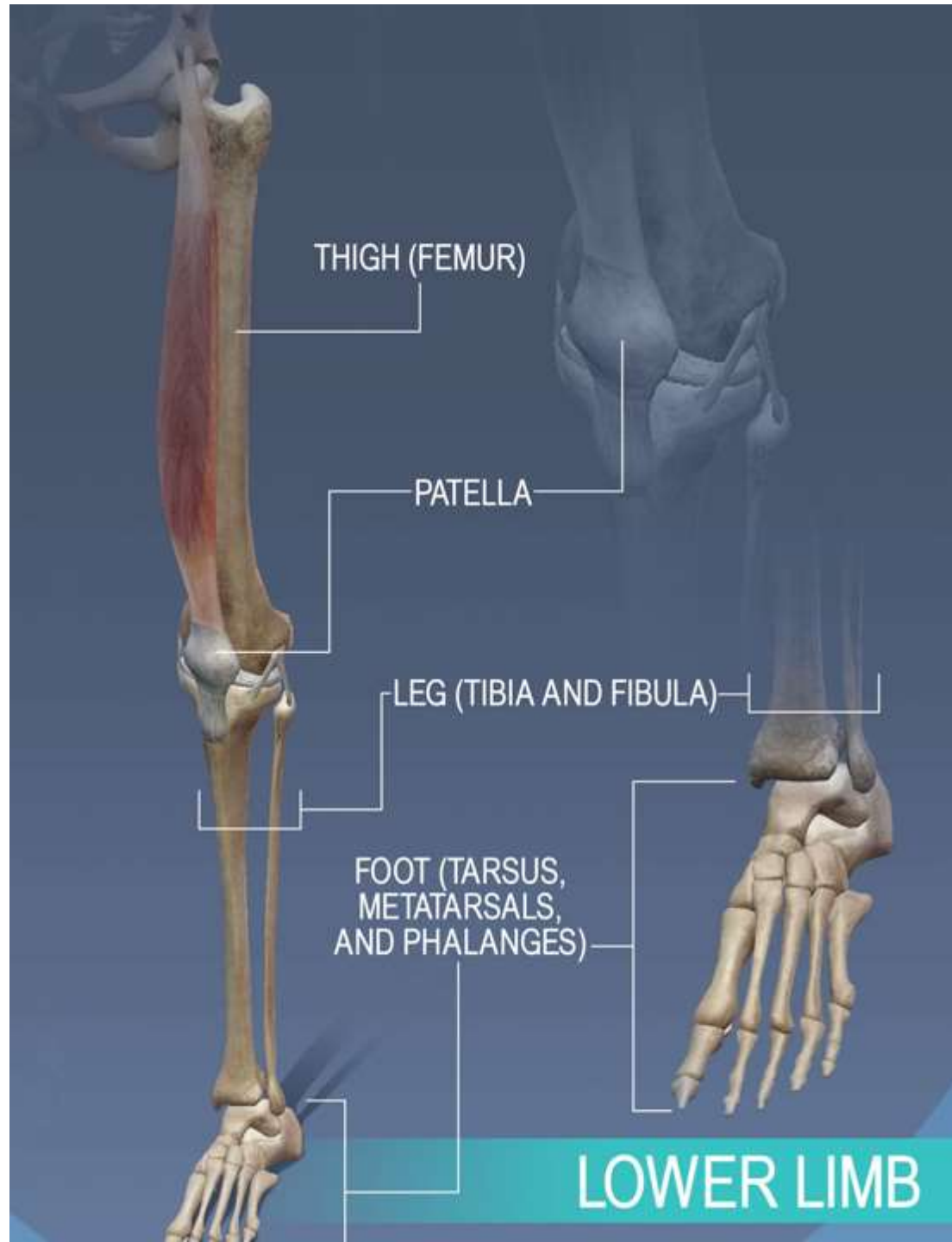


Appendicular Skeleton

Bones of the lower limb

By:

Dr. Nali Abdulkader



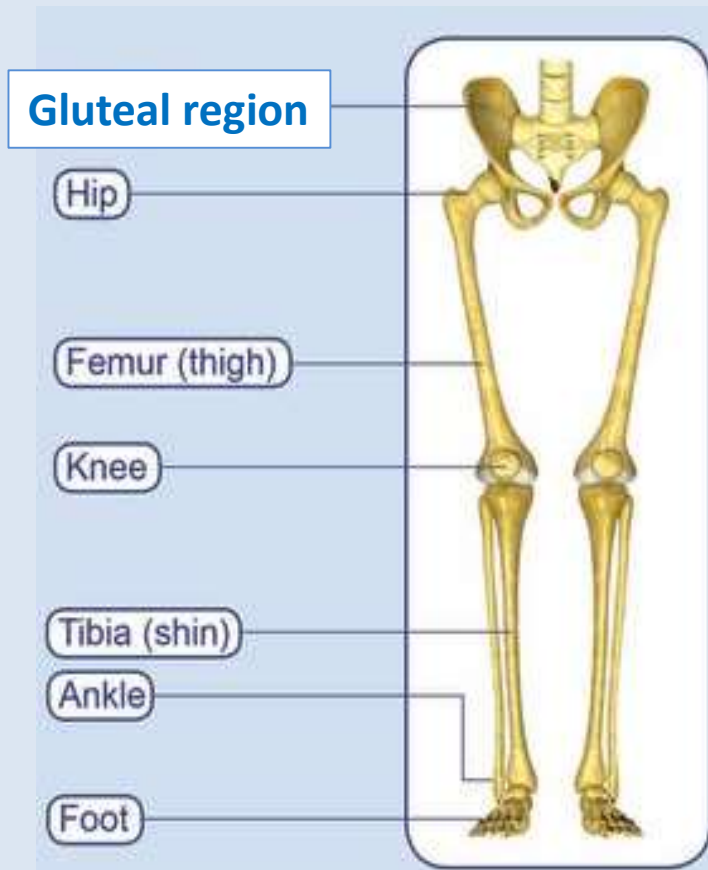
Parts of the lower limb

❖ The lower limb located inferiorly

❖ Functions are carry weight, supporting, locomotion

❖ The lower limb is divided into four parts

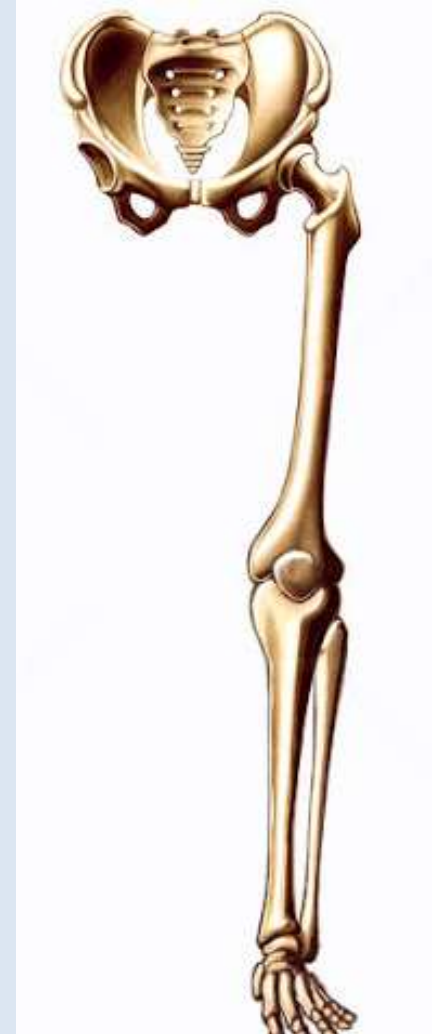
1. Gluteal region
2. Thigh
3. Leg
4. foot



Bones of the lower limb

❖ The bones of the lower limbs consists of 62 bones

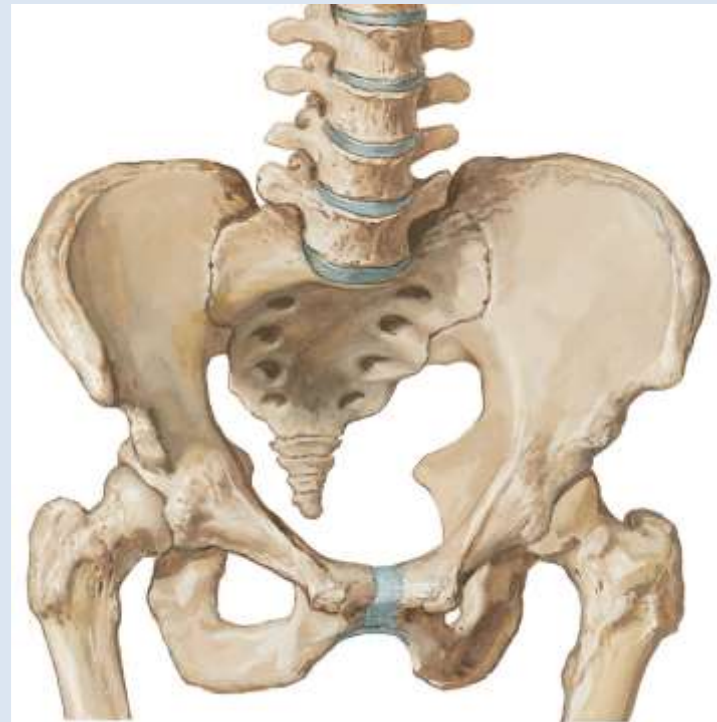
1. Bones of the pelvic girdle
2. Bone of the thigh
3. Patella
4. Bones of the leg and
5. Bones of the foot.



Components of the Bony Pelvis

❖ Bones of the pelvis composed four bones

- *Two hip bones*
- *Sacrum and*
- *Coccyx*



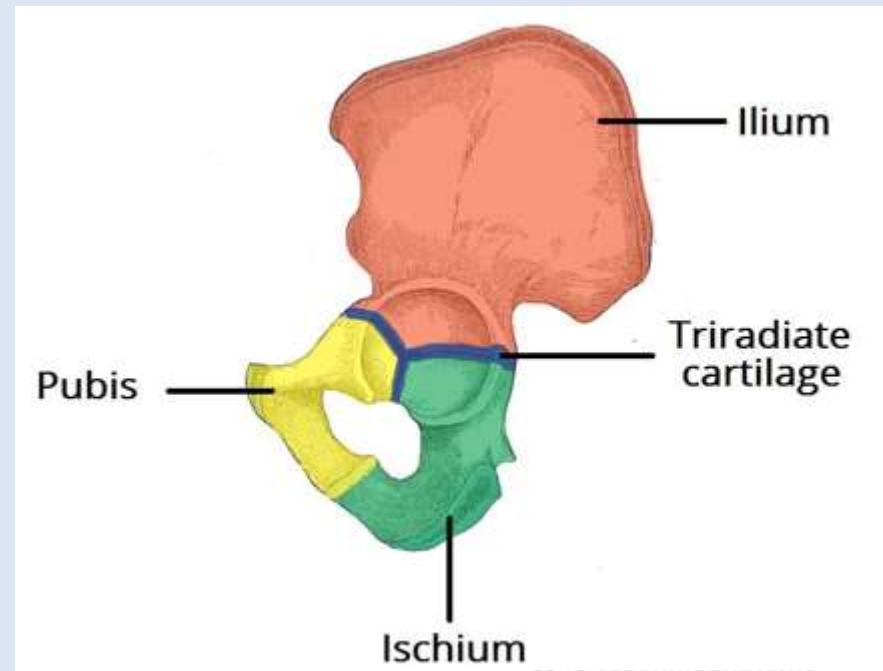
FUNCTIONS OF BONY PELVIS

- 1) To protect pelvic viscera
- 2) To support the weight of the body - transfer the weight of the upper body from the axial to the lower appendicular skeleton
- 3) Provides attachment for muscles
- 4) In females, it provide bony support for the birth canal



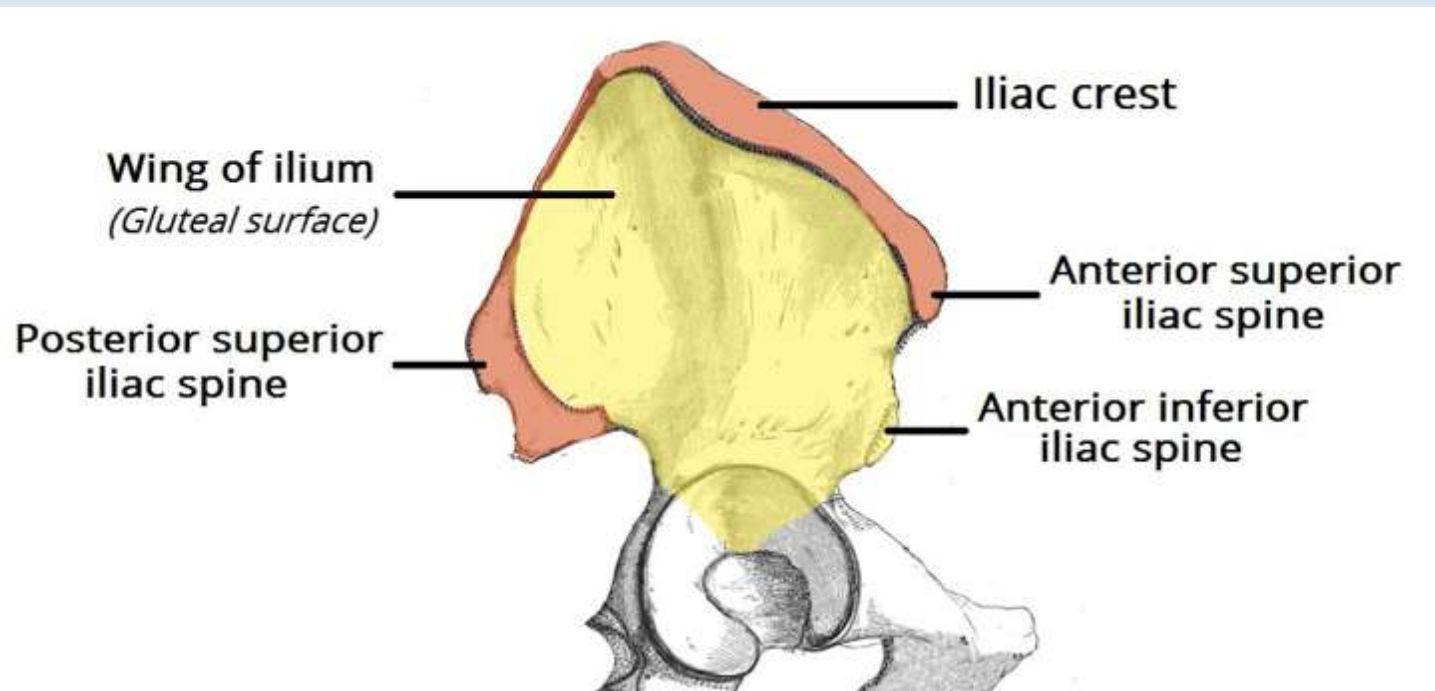
Hip bone

- ❑ Is a large flat or irregular bone formed by three elements **ilium, ischium and pubis**
- ❑ At the birth these bones are connected by cartilage at acetabulum called **triradiate cartilage**



Iliac crest

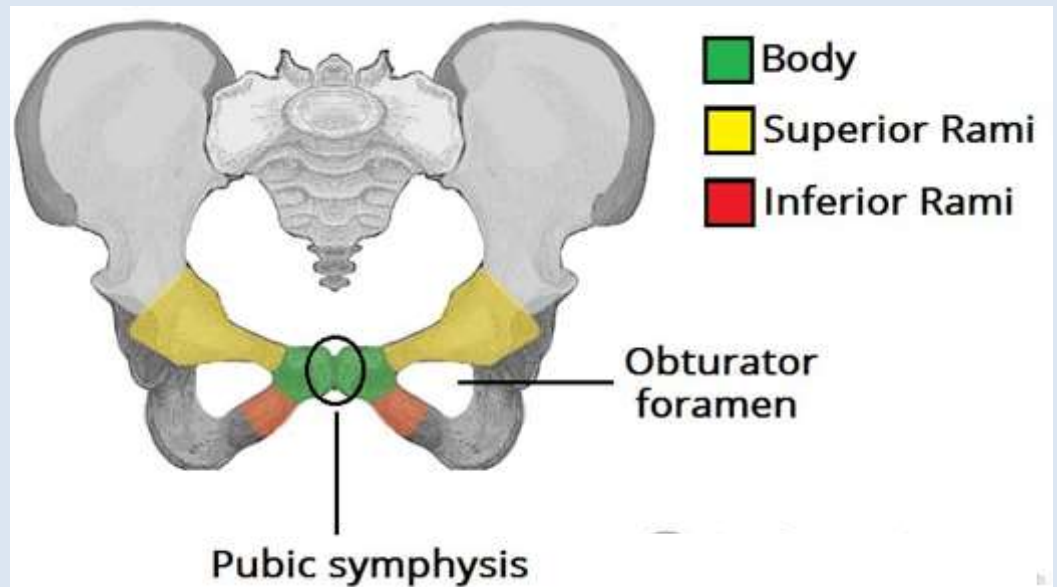
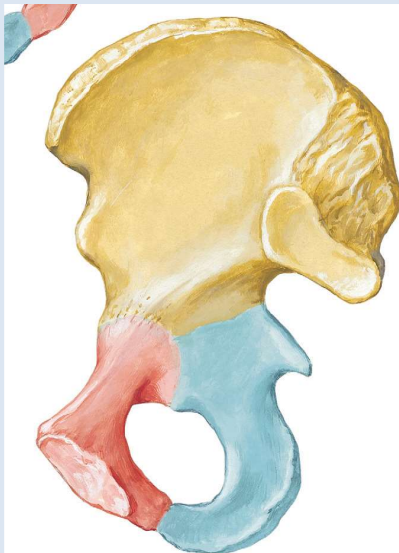
- Superior margin of the ilium is thickened to form iliac crest for attachment of muscles and fascia
- *Is the bone is frequently used for bone marrow biopsy*



Pubis

- **The pubis has:**

1. **Body**
2. **Superior ramus:**
3. **Inferior ramus:** forming ischiopubic rami

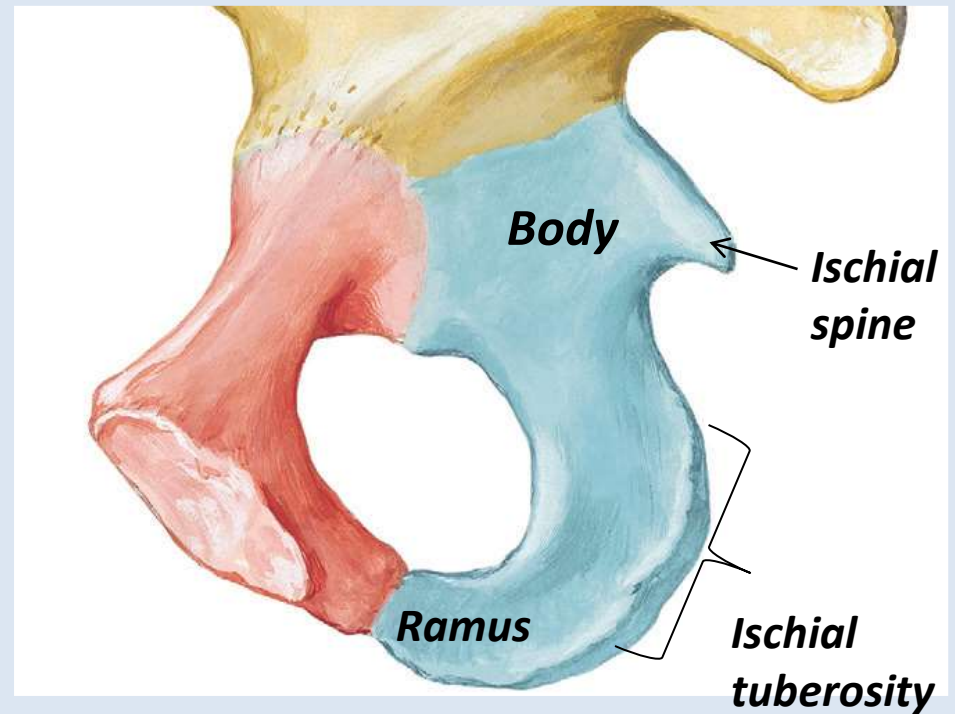


Ischium

❖ The ischium has

➤ *Body* it has

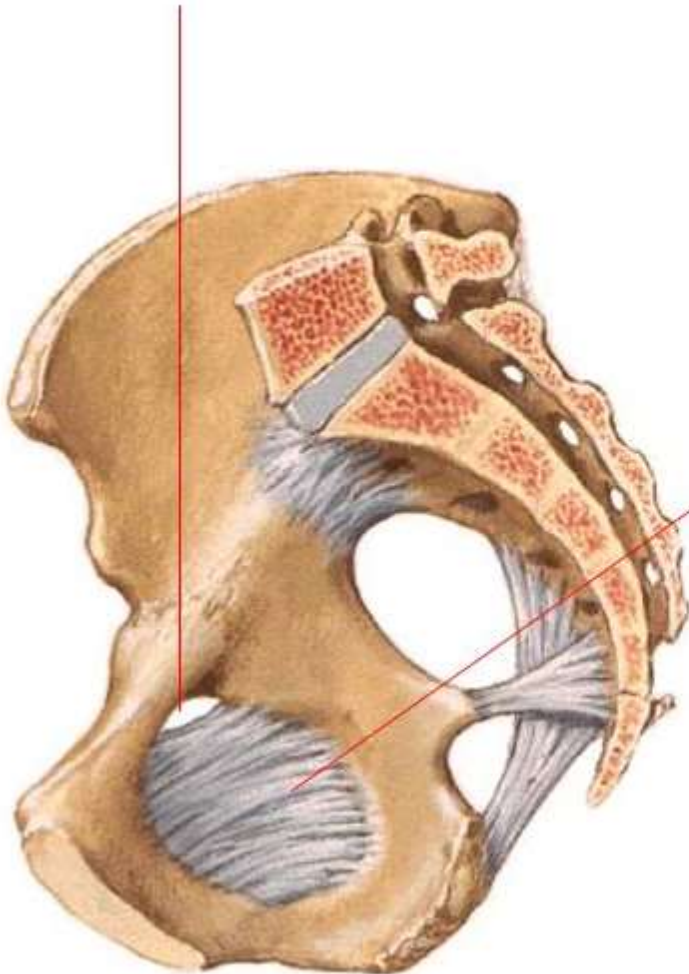
1. *Ischial spine*
2. *Ischial tuberosity*



The **obturator membrane** is a fibrous sheet that almost completely closes the obturator foramen, leaving a small gap, the **obturator canal**, for the passage of the obturator nerve and vessels as they leave the pelvis to enter the thigh.

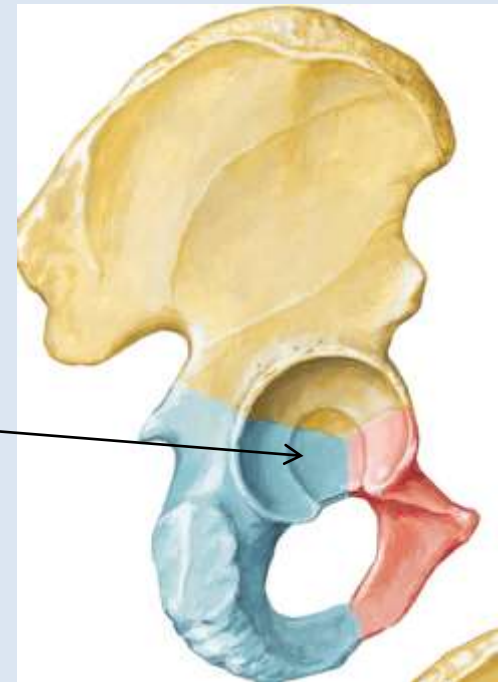
Obturator canal

Obturator membrane



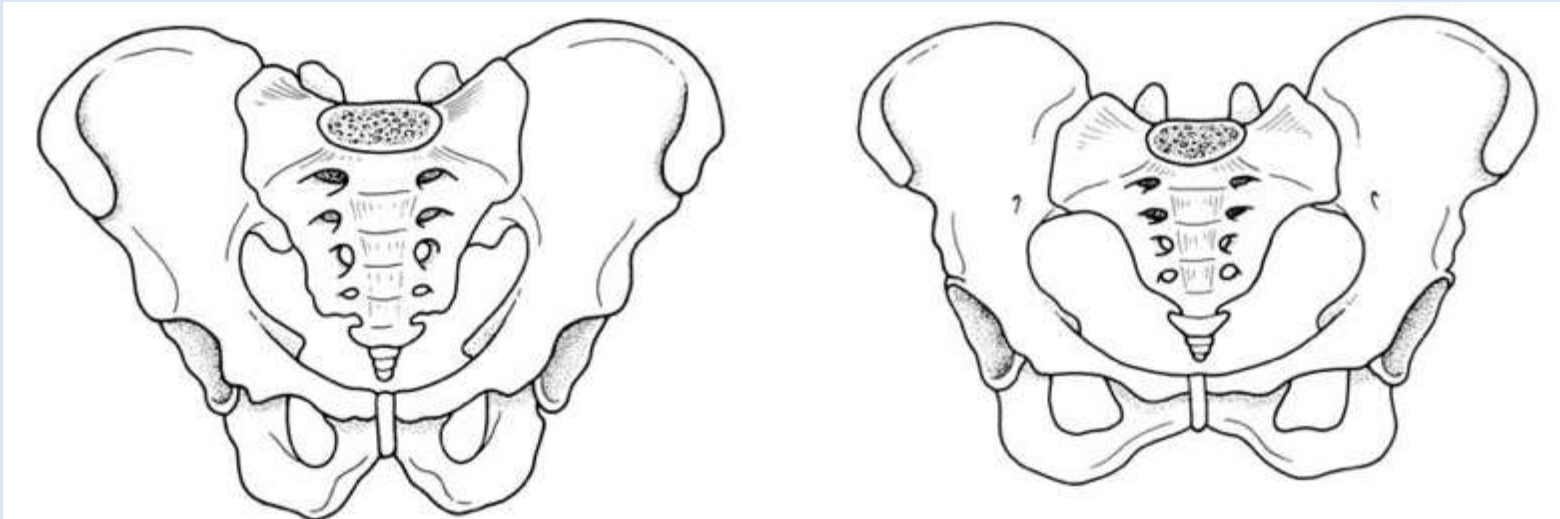
Acetabulum

- ❖ Is the cup shaped cavity on the lateral aspect of the hip bone
- ❖ It received the head of femur to form hip joint



Sex difference of the pelvis

- Pelvic inlet is oval in female and heart shape in male due to sacral promontory in male
- The ischial spines don't project medially in to the pelvis in female as they do in male
- The pubic angle is 80-90 in female while is 50-60 in male



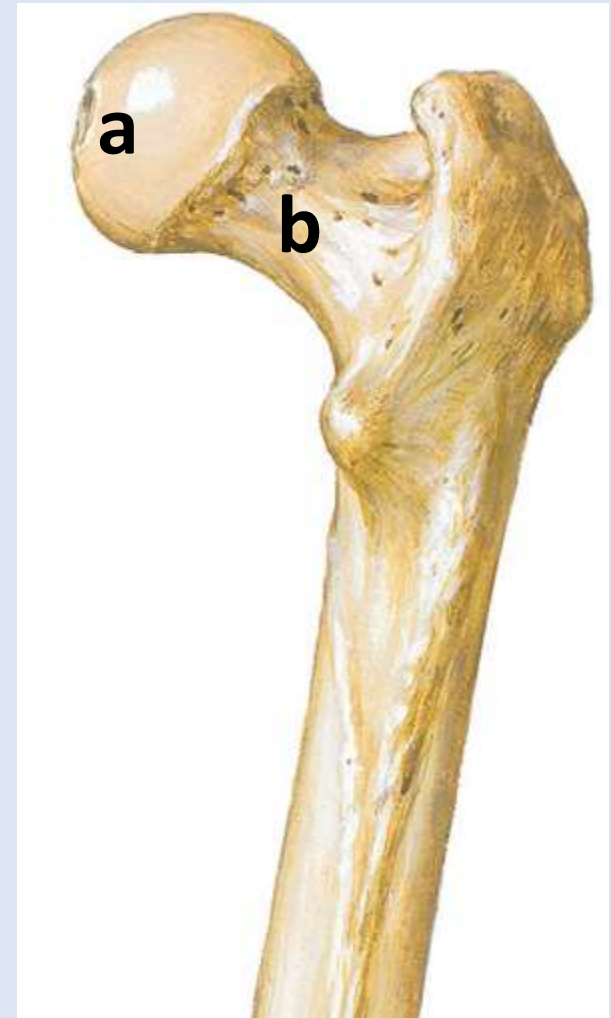
femur

- ❖ Is the largest bone of the body
Its length about a quarter
of the persons height
- ❖ It has upper and lower ends
and the shaft



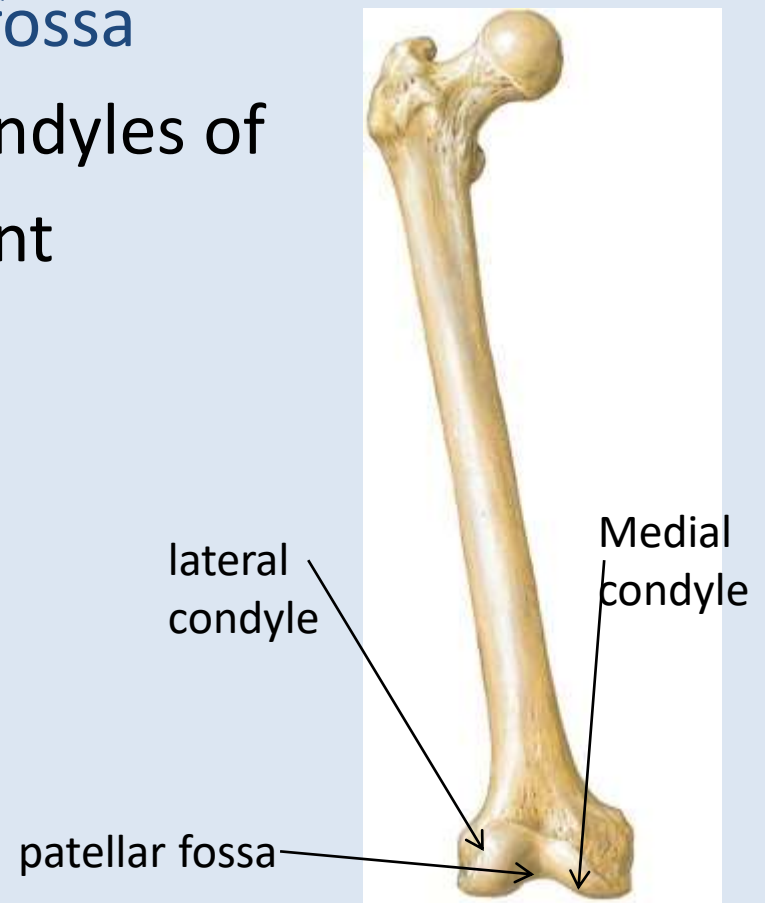
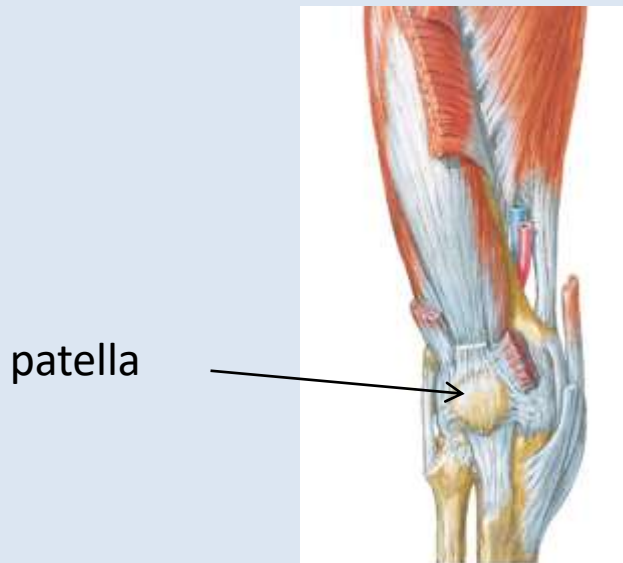
The upper end of femur

- ❖ It has a. Head and b. Neck
- ❖ **The head** articulates with the Acetabulum to form hip joint
- ❖ **Neck** is long and project medially and superiorly



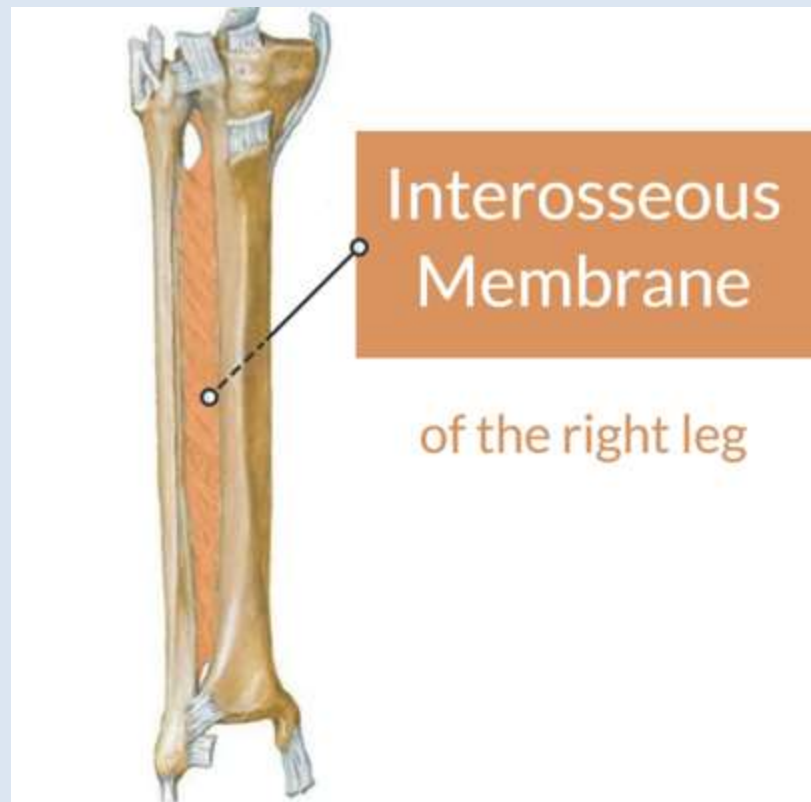
Distal end of femur

- The lower end of femur has **two condyles** separates from each other by **patellar fossa**
- The two condyles for two condyles of tibia to forming the knee joint



bones of the leg

- The tibia and fibula are the bones of the leg.
- The bones articulate with each other by interosseous membrane



Tibia

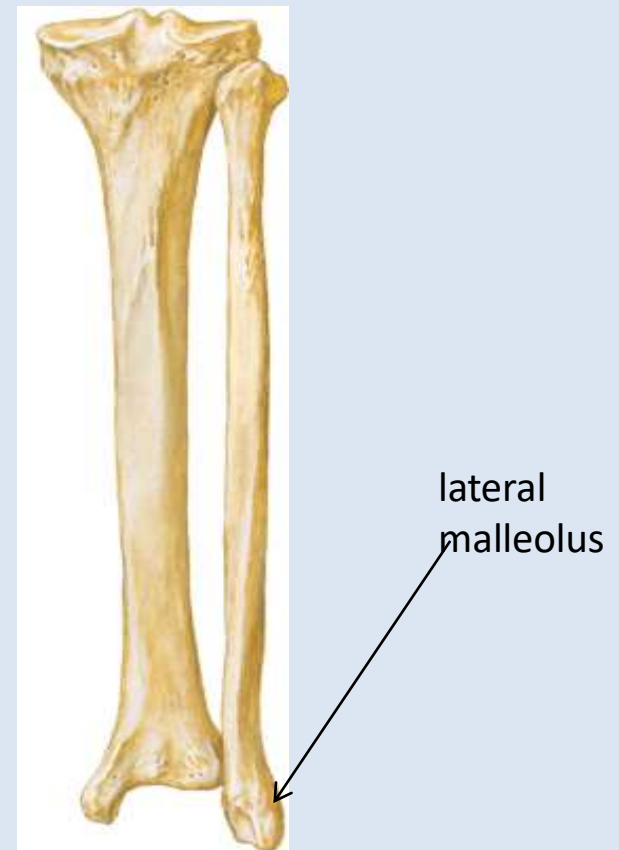
- ❖ The is the medial largest bone of the leg
- ❖ The tibia articulates with the condyles of the femur superiorly to form Knee joint and the talus inferiorly to forms ankle joint and in so doing transmits the body's weight



Fibula

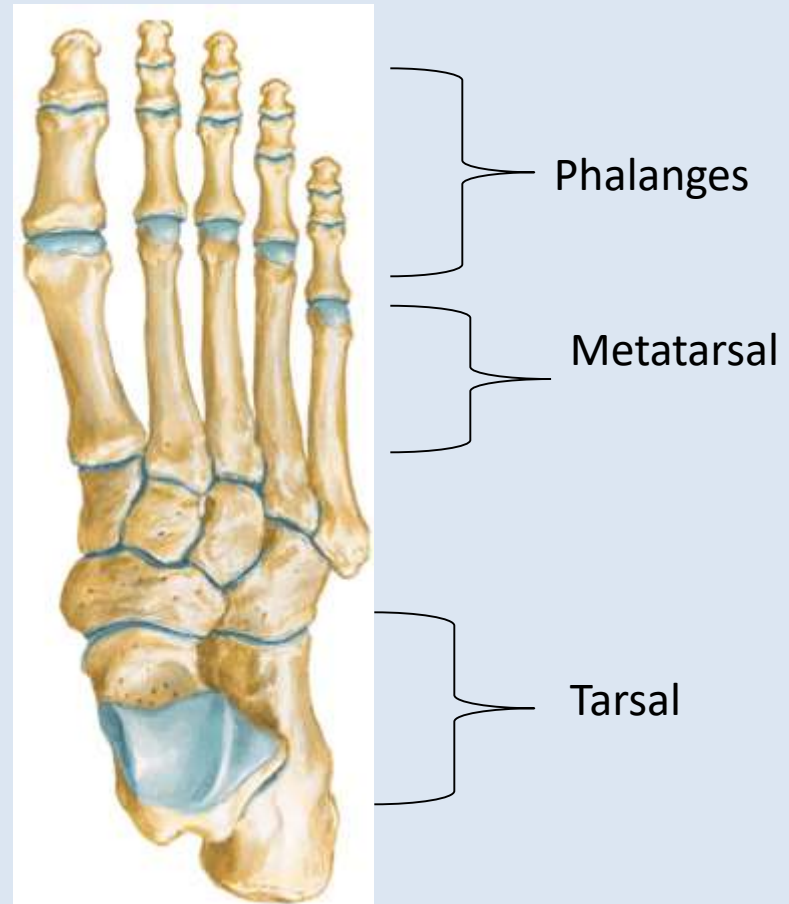
- The fibula mainly functions as an attachment for muscles, but it is also important for the stability of the ankle joint.

- Parallel and lateral to the tibia
- Smaller than the tibia
- Not weight bearing
- Not part of the knee joint
- Muscle attachments only
- Head on proximal end, lateral malleolus at ankle
- Fits into fibular notch at distal end of tibia



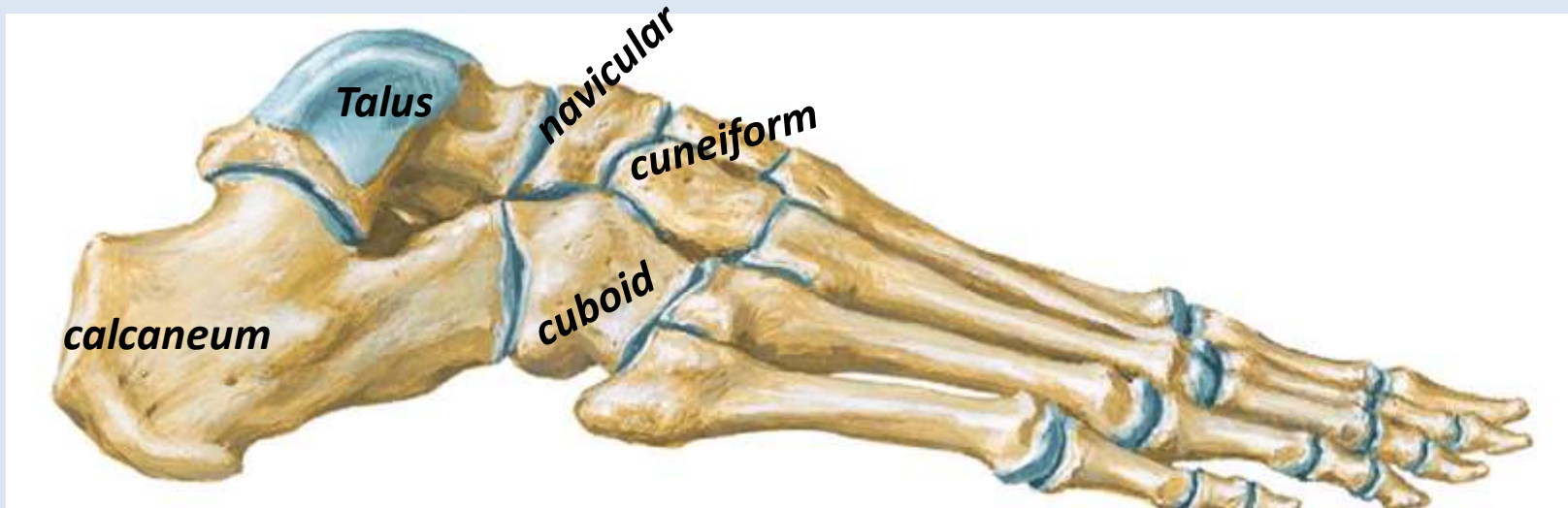
Bones of Foot

- The bones of the foot include the tarsus, metatarsus, and phalanges.
- *There are 7 tarsal bones*
- *5 metatarsal bones, and*
- *14 phalanges*



Tarsal bones

- There are seven tarsal bones which arranged in three groups
- Proximal group which formed by talus , calcaneum and navicular
- Distal group which is formed by cuboid and three cuneiform bones



How to learn the 7 tarsal names?

- Talus
- Calcaneus
- Navicular
- Cuneiforms: medial, intermediate, lateral
- Cuboid

Mnemonic for distal row is MILC:

medial, intermediate, lateral cuneiforms + Cuboid

Mnemonic for proximal 3 bones is: No Thanks Cow

That is Navicular, Talus, Calcaneus

Any
Question?

